

**Modeling Heterogeneous Headache
Frequency
Using Poisson Finite Mixture Models:**
An Application in R and SAS

Christina Agbor Abunaw

Master of Statistics (Biostatistics)

Transnational University Limburg (UHasselt), Belgium

Abstract

Headache frequency is a count outcome, but patients may not experience headaches at the same rate. When this type of heterogeneity is present, a standard Poisson model may not provide an adequate description of the data. This study examines the use of Poisson finite mixture models to identify underlying groups of patients with different headache frequencies.

The analysis used data from 401 patients, with the number of headache days during the baseline period as the main outcome and age considered as a covariate. The analysis included data preparation, descriptive statistics, assessment of over-dispersion, Poisson modelling, finite mixture modelling, nonparametric maximum likelihood estimation (NPMLE), and posterior classification. The analyses were carried out in both R and SAS.

The mean headache frequency was 16.18 days, compared with a variance of 44.98, giving a variance-to-mean ratio of 2.78. This indicated substantial over-dispersion and provided a reason to consider a mixture model rather than a single Poisson distribution. The NPMLE gradient analysis also indicated the presence of more than one underlying frequency group. Among the models considered, the three-component model had the lowest AIC, while BIC favored the two-component model. The three-component model was retained for further interpretation because it provided the lowest AIC and produced clearly distinguishable groups. The estimated mean headache frequencies for these groups were 9.61, 13.80, and 22.95 days.

Posterior classification showed that the patients could be separated into low-, moderate-, and high-frequency groups. The component means estimated using CAMAN and FlexMix were also similar, giving comparable results across the two approaches. When age was included in the three-component model, it was significantly associated with headache frequency in one component but not in the other two. Although the mixture model captured the main pattern of heterogeneity, it underestimated the number of patients reporting 28 headache days.

The results suggest that Poisson finite mixture models can provide a useful way of describing heterogeneous headache-frequency data when the assumptions of a standard Poisson model are not met. They also show the value of combining model diagnostics, mixture modelling, and posterior classification when studying variation in count outcomes.

Keywords: Poisson finite mixture model; headache frequency; over-dispersion; latent groups; NPMLE; posterior classification; R; SAS

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1 Introduction

1.1 Background

Headache disorders are common health conditions that can substantially affect quality of life, daily functioning, and healthcare utilization. In clinical studies, headache burden is often measured using count outcomes, such as the number of days on which a patient experiences a headache during a specified observation period.

The Poisson distribution is commonly used to model count outcomes because it provides a natural probabilistic framework for the occurrence of events. A conventional Poisson model, however, assumes that the conditional mean and variance of the outcome are equal. This assumption may be inappropriate when patients differ substantially in their underlying susceptibility to headaches.

When observations arise from a population containing individuals with different underlying headache frequencies, the resulting data may exhibit latent heterogeneity. One consequence of such heterogeneity can be over-dispersion, whereby the observed variance exceeds the mean. Ignoring this structure and applying a single-population Poisson model may therefore result in an inadequate representation of the observed count distribution.

Finite mixture models provide a flexible framework for modelling such latent heterogeneity. Rather than assuming that all observations arise from a single population, a finite mixture model assumes that the population consists of a finite number of unobserved subpopulations, or mixture components. A Poisson finite mixture model (PFMM) allows each component to have its own Poisson rate while simultaneously estimating the proportion of individuals belonging to each latent component.

This study applies Poisson finite mixture modelling to headache frequency data from a randomized clinical trial. The analysis considers the distribution of headache frequency, evaluates evidence of over-dispersion, estimates Poisson finite mixture models, and identifies an appropriate number of latent components. Nonparametric maximum likelihood estimation (NPMLE) and the expectation-maximization (EM) algorithm are used in the estimation process, while posterior probabilities are used to classify patients into the identified latent subpopulations.

Patient age is additionally considered as a covariate to investigate whether an observed patient characteristic can explain differences in headache frequency across the latent mixture components. The analysis is implemented using both R and SAS, providing a reproducible computational framework for the statistical investigation.

1.2 Problem Statement

Headache frequency may vary considerably between patients, even when the patients are enrolled in the same clinical study. Such variation may reflect underlying differences

in susceptibility to headaches and may result in a population that is not adequately described by a single probability distribution.

A standard Poisson model assumes that all observations arise from one population with a common underlying event rate, subject to the model's covariate structure. When the population contains unobserved subgroups with different event rates, this assumption may be violated and the resulting data may exhibit over-dispersion.

The statistical problem addressed in this study is therefore to determine whether the observed heterogeneity in headache frequency can be represented by a finite mixture of Poisson distributions and to identify the number of latent components that provides an appropriate representation of the data.

A related objective is to investigate whether patient age helps explain the differences in headache frequency associated with the identified mixture components. This provides an opportunity to distinguish heterogeneity that can be explained by an observed covariate from heterogeneity that may remain unexplained by the available data.

1.3 Research Objectives

1.3.1 General Objective

The general objective of this study is to investigate latent heterogeneity in headache frequency using Poisson finite mixture models and to evaluate their ability to represent over-dispersed count data.

1.3.2 Specific Objectives

The specific objectives are to:

1. Describe the distribution and descriptive characteristics of headache frequency in the study population.
2. Assess whether the observed headache frequency data exhibit over-dispersion relative to a conventional Poisson model.
3. Estimate Poisson finite mixture models with different numbers of latent components.
4. Compare the fitted mixture models and identify an appropriate number of latent subpopulations.
5. Estimate the parameters of the selected mixture model using nonparametric maximum likelihood estimation and the expectation-maximization algorithm.
6. Classify patients into the identified latent subpopulations using posterior probabilities.

7. Investigate whether patient age explains differences in headache frequency across the identified mixture components.
8. Implement the statistical analysis using both R and SAS.

1.4 Research Questions

The study addresses the following research questions:

1. What is the distribution of headache frequency among the patients in the study population?
2. Does the headache frequency data exhibit over-dispersion relative to a single-population Poisson model?
3. How many latent subpopulations provide an appropriate representation of the observed headache frequency distribution?
4. What are the estimated Poisson rates and mixing proportions of the selected finite mixture model?
5. How can patients be classified into the identified latent subpopulations using posterior probabilities?
6. Does patient age help explain differences in headache frequency across the latent mixture components?

1.5 Scope of the Study

This study focuses on the statistical modelling of headache frequency recorded during a four-week baseline period. The primary outcome is the number of headache days, treated as a count variable.

The analysis focuses on Poisson-based finite mixture modelling, including assessment of over-dispersion, estimation and comparison of mixture models, latent subpopulation classification, and the incorporation of patient age as a covariate.

The study is concerned with statistical description and inference rather than the evaluation of treatment effectiveness. Consequently, the analysis does not attempt to establish causal relationships between patient characteristics and headache frequency.

The computational analysis is conducted using R and SAS, while L^AT_EX is used to prepare and document the research report.

1.6 Structure of the Report

The remainder of this report is organized as follows.

Section 2 describes the study data and presents the exploratory and descriptive analyses.

Section 3 presents the literature review, covering count-data modelling, heterogeneity and finite mixture models, nonparametric maximum likelihood estimation, estimation of finite mixture models, and approaches to determining the number of components.

Section 4 presents the statistical methodology, including the Poisson distribution, assessment of over-dispersion, Poisson finite mixture models, nonparametric maximum likelihood estimation, the expectation-maximization algorithm, and posterior probability-based classification.

Section 5 presents the model estimation, model comparison, and principal statistical results.

Section 6 discusses the findings in relation to the research objectives, including the role of patient age and the implications of the identified latent heterogeneity.

Section 7 summarizes the main findings and presents the conclusions and implications of the study.

2 Data and Exploratory Analysis

2.1 Study Data

The data used in this analysis were obtained from a randomized clinical trial comparing acupuncture and placebo treatments for chronic headache [9]. The dataset contains 401 patients.

The primary outcome is the number of headache days recorded during the four-week baseline period. Patient age at baseline was also available and was considered as a covariate in the subsequent mixture analysis.

The purpose of the exploratory analysis was to examine the structure of the data before fitting the statistical models, with particular attention to the distribution and variability of headache frequency.

2.2 Variables

Two variables were used in the analysis.

Table 1: Variables included in the analysis

Variable	Type	Description
Frequency	Count	Number of headache days during the four-week baseline period
Age	Continuous	Patient age at baseline

Frequency is the response variable used in the Poisson models. Since it represents a number of headache days, it is treated as a count outcome. Age is included later as a covariate in the three-component mixture model.

2.3 Data Preparation

The original headache-frequency variable contained five non-integer observations. Since the Poisson distribution is defined for integer-valued counts, these observations were rounded to the nearest whole number before the statistical models were fitted.

The data were checked for missing values, variable types, and plausible ranges before analysis. The final cleaned dataset was then used consistently in both the R and SAS analyses.

The individual non-integer observations and the details of the data cleaning procedure are reported in the Results section.

2.4 Descriptive Analysis

The distribution of baseline headache frequency was examined using summary statistics, a frequency distribution, and graphical diagnostics. Measures of central tendency and variability were used to describe the overall level of headache frequency, while the histogram was used to examine the shape of the observed distribution.

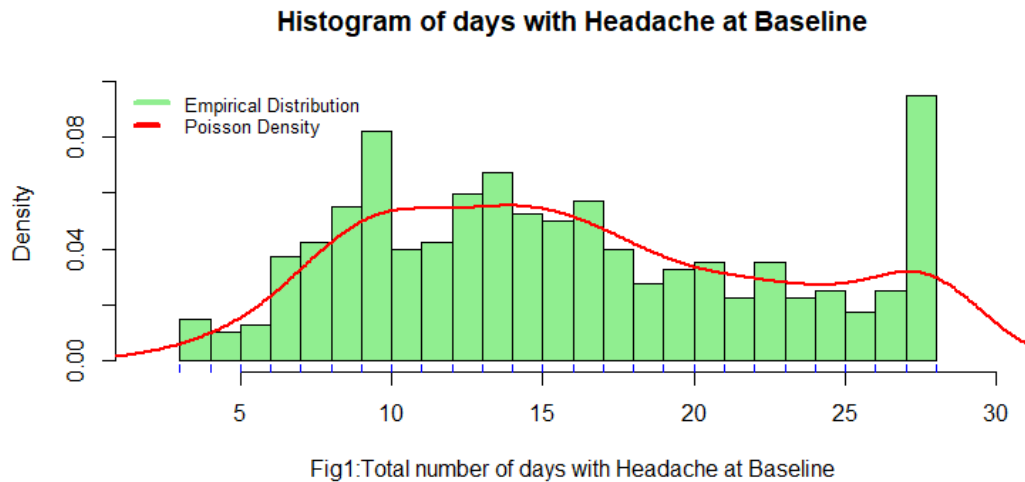


Figure 1: Distribution of headache days at baseline. The bars show the empirical distribution and the curve represents the fitted Poisson density based on the sample mean.

The distribution shows considerable variation in the number of headache days between patients. In particular, the observations are not concentrated around a single narrow range, and a relatively high frequency is observed near the upper boundary of 28 headache days.

These features suggest that a single Poisson distribution may not fully describe the observed data.

2.5 Initial Assessment of Poisson Assumptions

The Poisson model assumes that the mean and variance of the outcome are equal. Therefore, comparing the observed mean and variance provides an initial assessment of whether a standard Poisson model is appropriate.

The exploratory analysis showed that the variance of headache frequency was substantially larger than its mean. This indicates over-dispersion and suggests that patients may differ in their underlying headache-frequency rates.

Rather than treating this excess variation only as a violation of the Poisson assumption, the analysis considers whether it may reflect latent heterogeneity within the patient population. A finite mixture of Poisson distributions provides a natural framework for examining this possibility.

The formal dispersion results and subsequent model estimates are presented in the Results section.

2.6 Exploratory Modelling Strategy

The exploratory findings guided the subsequent modelling strategy. A single-component Poisson model was first used as a reference. Finite mixture models were then considered to determine whether the observed heterogeneity could be represented by several latent headache-frequency groups.

NPMLE was used as an exploratory tool to examine the underlying mixing distribution before selecting a finite number of components. The final mixture models were then compared using information criteria and model diagnostics.

This approach allowed the analysis to move from a description of the observed data to an assessment of whether latent heterogeneity could provide a better representation of headache frequency.

3 Literature Review

3.1 Count Data and the Poisson Distribution

Count outcomes occur frequently in medical and epidemiological research. Examples include the number of hospital visits, disease episodes, adverse events, and days with symptoms. The Poisson distribution is one of the standard probability models for such outcomes because it provides a simple representation of event counts through a single rate parameter.

A key property of the Poisson distribution is equidispersion, meaning that the mean and variance are equal. This assumption can be restrictive in applications where individuals differ in their underlying event rates. When the observed variance is substantially larger than the mean, a standard Poisson model may therefore provide a poor description of the data. The assessment of such assumptions forms an important part of count-data modelling [6].

Over-dispersion can arise for several reasons, including unobserved heterogeneity between individuals, dependence between events, and model misspecification. Alternative count-data models have therefore been developed to accommodate departures from the standard Poisson assumption, including negative binomial and COM-Poisson models [8, 2].

For the present study, the main interest is not simply to allow for over-dispersion but to investigate whether the excess variation in headache frequency may reflect distinct underlying groups of patients. This motivates the use of finite mixture models.

3.2 Heterogeneity and Finite Mixture Models

Finite mixture models provide a way of representing a population as a combination of several unobserved component distributions. Instead of assuming that all observations arise from the same probability distribution, the population is assumed to contain a finite number of latent groups, each with its own parameters and mixing proportion [5].

For count data, a Poisson finite mixture model assumes that the observed counts arise from one of several Poisson distributions. The latent component is not observed directly, but the probability that an observation belongs to each component can be estimated from the data.

This approach is useful when over-dispersion is thought to result from differences between subpopulations. A mixture of Poisson distributions can have a variance greater than its mean even though each individual component follows a Poisson distribution. The additional variation arises from differences between the component-specific Poisson means.

Finite mixture models have been applied in a wide range of areas, including medical

research, where populations may contain individuals with different underlying disease or risk profiles [5]. In the context of headache frequency, this provides a useful framework for investigating whether patients can be described by different underlying levels of headache burden rather than by a single population-average rate.

3.3 Nonparametric Maximum Likelihood Estimation

Before specifying a fixed number of mixture components, the underlying mixing distribution can be investigated using nonparametric maximum likelihood estimation (NPMLE). Rather than assuming in advance that the mixing distribution has a particular number of support points, NPMLE estimates the distribution directly from the observed data.

The theoretical basis for NPMLE in mixture models was developed by Lindsay, who examined the geometry of mixture likelihoods and the properties of estimated mixing distributions [3, 4].

NPMLE is useful in the present analysis because it provides an exploratory assessment of whether the data support a homogeneous population or several underlying support points. The gradient function can then be examined to identify regions that may correspond to candidate mixture components.

This provides a link between exploratory analysis and the subsequent finite mixture models. Rather than choosing the number of components only by repeatedly fitting increasingly complex models, the NPMLE analysis provides information about the possible structure of the underlying mixing distribution.

3.4 Estimation of Finite Mixture Models

Finite mixture models contain an unobserved component-membership variable, which makes direct maximum likelihood estimation more complicated. The expectation-maximization (EM) algorithm provides a general approach for estimating models with this type of missing or unobserved information [5].

The EM algorithm alternates between two stages. In the expectation step, the current parameter estimates are used to calculate the posterior probability that each observation belongs to each mixture component. In the maximization step, the parameters are updated using these probabilities. The process is repeated until convergence.

The EM framework is particularly relevant to the present study because component membership is not directly observed. The estimated posterior probabilities provide both a means of estimating the model parameters and a basis for classifying individual patients according to their most likely latent headache-frequency group.

3.5 Determining the Number of Components

An important issue in finite mixture modelling is determining how many components should be included. Adding components generally improves the likelihood, but excessive complexity can result in overfitting. Model selection therefore requires consideration of both goodness of fit and model complexity [5].

Information criteria provide a practical approach to this problem. The Akaike Information Criterion (AIC), introduced by Akaike [1], balances model fit against the number of estimated parameters. Smaller AIC values indicate a preferred trade-off between fit and complexity.

The Bayesian Information Criterion (BIC), proposed by Schwarz [7], applies a stronger penalty for model complexity than AIC. It is therefore possible for AIC and BIC to select different numbers of components when candidate models have similar likelihoods but different numbers of parameters.

In finite mixture analysis, component selection should therefore not depend on a single numerical criterion. Information criteria can be considered alongside likelihood values, diagnostic plots, estimated mixing proportions, and the interpretability of the resulting components [5].

This consideration is particularly relevant to the present study, where the two- and three-component models provide different results under AIC and BIC.

3.6 Mixture Models and Headache Frequency

Headache frequency is naturally represented as a count outcome when the outcome is defined as the number of headache days during a fixed period. The clinical data used in this study originate from a randomized trial investigating acupuncture for chronic headache in primary care [9].

A conventional Poisson model provides an initial framework for such data, but it assumes that patients share a common underlying rate after accounting for included covariates. This assumption may be unrealistic when there are substantial differences in headache frequency between patients.

Patients with relatively few headache days may have a different underlying rate from patients who experience headaches on most days of the observation period. A finite mixture model provides a way to represent this possibility without requiring the latent groups to be defined in advance.

The latent components are estimated from the observed distribution and can subsequently be examined for their statistical and clinical interpretability. The present study therefore uses Poisson finite mixture modelling to investigate whether the heterogeneity in baseline headache frequency can be represented by a small number of latent groups.

3.7 Summary of the Literature

The literature provides a statistical basis for the modelling strategy used in this study. The Poisson distribution provides a natural starting point for count outcomes, but its equidispersion assumption can be restrictive when substantial heterogeneity is present. Finite mixture models provide one approach to representing this heterogeneity through latent subpopulations [6, 5].

NPMLE provides a flexible approach for exploring the underlying mixing distribution, while the EM algorithm provides a practical method for estimating finite mixture parameters [3, 5]. AIC and BIC provide complementary criteria for comparing models with different numbers of components [1, 7].

Together, these methods provide the framework used in this study to investigate the distribution of headache frequency and to determine whether a finite mixture model provides a more informative description than a single Poisson distribution.

4 Statistical Methodology

4.1 Overview

The analysis was designed to investigate whether the observed heterogeneity in baseline headache frequency could be represented by a finite number of latent Poisson distributions. The analysis was conducted in several stages. The data were first inspected and cleaned, followed by descriptive analysis and assessment of over-dispersion. A single-component Poisson model was then fitted as a reference model. Poisson finite mixture models with different numbers of components were subsequently estimated and compared.

Nonparametric maximum likelihood estimation (NPMLE) was used as an exploratory approach to investigate the underlying mixing distribution. The expectation-maximization (EM) framework was used in the estimation of finite mixture models. Posterior probabilities were used to classify patients according to their most likely latent component.

The analyses were implemented using both R and SAS. R was used particularly for NPMLE analysis, gradient-function assessment, FlexMix modelling, posterior classification, and graphical presentation. SAS was used for the Poisson reference model, finite mixture model estimation, model comparison, and the analysis including age as a covariate.

4.2 Poisson Reference Model

Let Y_i denote the number of headache days observed for patient i , where $i = 1, \dots, n$. Under the standard Poisson model,

$$Y_i \sim \text{Poisson}(\lambda),$$

where $\lambda > 0$ represents the expected number of headache days.

The probability mass function is

$$P(Y_i = y_i) = \frac{\exp(-\lambda)\lambda^{y_i}}{y_i!}, \quad y_i = 0, 1, 2, \dots$$

For a Poisson distribution,

$$E(Y_i) = \lambda$$

and

$$\text{Var}(Y_i) = \lambda.$$

Consequently, equality between the mean and variance is an important diagnostic assumption of the standard Poisson model.

4.3 Assessment of Over-dispersion

The observed variance was compared with the observed mean using the variance-to-mean ratio,

$$D = \frac{\text{Var}(Y)}{E(Y)}.$$

Values substantially greater than one indicate over-dispersion relative to the Poisson assumption.

In this study, the dispersion ratio was calculated from the cleaned headache-frequency variable before fitting the finite mixture models.

4.4 Poisson Finite Mixture Model

To account for potential heterogeneity between patients, a finite mixture of Poisson distributions was considered.

For a model with K components,

$$P(Y_i = y_i) = \sum_{k=1}^K \pi_k \frac{\exp(-\lambda_k) \lambda_k^{y_i}}{y_i!},$$

where λ_k is the Poisson mean for component k and π_k is the corresponding mixing proportion.

The mixing proportions satisfy

$$\pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

The marginal mean of the mixture is

$$E(Y) = \sum_{k=1}^K \pi_k \lambda_k.$$

The marginal variance is

$$\text{Var}(Y) = \sum_{k=1}^K \pi_k \lambda_k + \sum_{k=1}^K \pi_k (\lambda_k - \bar{\lambda})^2,$$

where

$$\bar{\lambda} = \sum_{k=1}^K \pi_k \lambda_k.$$

The second term represents additional variation arising from differences between the latent components. Thus, a Poisson mixture can produce over-dispersion even though

each individual component is Poisson.

4.5 Nonparametric Maximum Likelihood Estimation

NPMLE was used to explore the underlying mixing distribution without fixing the number of mixture components in advance. The mixing distribution was estimated using the CAMAN package in R.

The NPMLE analysis was used as an exploratory step rather than as the sole criterion for selecting the final finite mixture model. The estimated gradient function was examined to identify regions of the parameter space that could correspond to support points of the mixing distribution.

The NPMLE procedure was implemented using the VEM algorithm followed by EM refinement. The analysis used 50,000 maximum iterations, a convergence accuracy of 10^{-7} , and an initial support size of 25.

4.6 Expectation-Maximization Algorithm

Finite mixture models contain an unobserved component-membership variable. Let Z_i denote the latent component for patient i .

The EM algorithm alternates between two steps.

In the expectation step, the posterior probability that observation i belongs to component k is calculated as

$$\tau_{ik} = P(Z_i = k \mid Y_i) = \frac{\pi_k f_k(Y_i)}{\sum_{j=1}^K \pi_j f_j(Y_i)},$$

where $f_k(Y_i)$ denotes the Poisson probability under component k .

In the maximization step, the model parameters are updated using the posterior probabilities from the expectation step. The two steps are repeated until convergence.

The EM framework provides both parameter estimates and posterior component probabilities, which are subsequently used for patient classification.

4.7 Model Estimation and Component Selection

Poisson mixture models containing two, three, four, and five components were fitted. A single-component Poisson model was retained as the reference model.

The models were compared using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). These criteria balance goodness of fit against model complexity.

The AIC is defined as

$$\text{AIC} = -2\ell(\hat{\theta}) + 2p,$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood and p is the number of estimated parameters.

The BIC is defined as

$$\text{BIC} = -2\ell(\hat{\theta}) + p \log(n),$$

where n is the sample size.

For both criteria, smaller values indicate a preferred model.

Because different information criteria can select different numbers of components, model selection was considered together with the NPMLE results, estimated component parameters, model diagnostics, and interpretability of the resulting latent groups.

4.8 Posterior Classification

After fitting the three-component model, each patient was assigned to the component with the largest posterior probability,

$$\hat{Z}_i = \arg \max_k \tau_{ik}.$$

This classification was used to visualize the distribution of patients across the estimated latent headache-frequency groups.

The classification was treated as a probabilistic model-based assignment rather than as a definitive clinical diagnosis.

4.9 Age-Adjusted Mixture Model

Age was subsequently included as a covariate in the three-component Poisson mixture model.

For component k , the Poisson mean was modelled using a log link,

$$\log(\lambda_{ik}) = \beta_{0k} + \beta_{1k} \text{Age}_i.$$

Equivalently,

$$\lambda_{ik} = \exp(\beta_{0k} + \beta_{1k} \text{Age}_i).$$

The coefficient β_{1k} represents the association between age and the expected headache frequency within component k on the logarithmic scale.

The statistical significance of the age effect was assessed using the parameter estimates and corresponding standard errors and p -values from the fitted SAS model.

4.10 Software and Reproducibility

All data preparation and statistical analyses were performed using reproducible scripts.

R was used for data preparation, descriptive analysis, NPMLE estimation, gradient-function analysis, finite mixture modelling, posterior classification, and graphical output.

SAS Studio was used independently to import and inspect the data, perform descriptive analysis, fit Poisson and finite mixture models, compare candidate models, and evaluate the final three-component model including age.

The R and SAS analyses were maintained as separate scripts so that the principal findings could be compared across software implementations.

5 Results

5.1 Data Preparation and Quality Assessment

The final analytical dataset contained 401 observations with information on age and baseline headache frequency. Headache frequency was treated as a count variable representing the number of headache days during the baseline period.

Initial inspection identified five observations with non-integer headache-frequency values. These observations were 10.67, 18.67, 26.67, 22.67, and 26.67 days. Because the Poisson distribution is defined for integer-valued counts, the five values were rounded to the nearest whole number before fitting the models. No missing observations were identified for age or headache frequency.

Following data preparation, headache frequency ranged from 3 to 28 days.

Table 2: Non-integer headache-frequency observations identified during data preparation

Age	Original Frequency
23	10.67
36	18.67
28	26.67
23	22.67
33	26.67

5.2 Descriptive Analysis

The mean baseline headache frequency was 16.18 days, with a standard deviation of 6.71 days. The median was 15 days, and the observed values ranged from 3 to 28 days.

Table 3: Descriptive statistics for baseline headache frequency

Statistic	Value
Sample size	401
Mean	16.18
Variance	44.98
Standard deviation	6.71
Median	15.00
Minimum	3.00
Maximum	28.00
Variance-to-mean ratio	2.78

The variance was considerably larger than the mean, indicating substantial variability in headache frequency between patients.

5.3 Assessment of Over-dispersion

Under a standard Poisson model, the mean and variance are expected to be equal. For the present data, the estimated mean was 16.18 and the estimated variance was 44.98. The resulting variance-to-mean ratio was

$$\frac{\text{Var}(Y)}{E(Y)} = \frac{44.98}{16.18} = 2.78.$$

The ratio was substantially greater than one, indicating over-dispersion relative to the standard Poisson assumption. This provided a clear reason to investigate models that could account for heterogeneity between patients.

5.4 NPMLE Gradient-Function Analysis

A nonparametric maximum likelihood estimate (NPMLE) was obtained using the CAMAN package as an exploratory analysis of the underlying mixing distribution. The NPMLE was estimated using the VEM algorithm followed by EM refinement.

The resulting gradient function is shown in Figure 2.

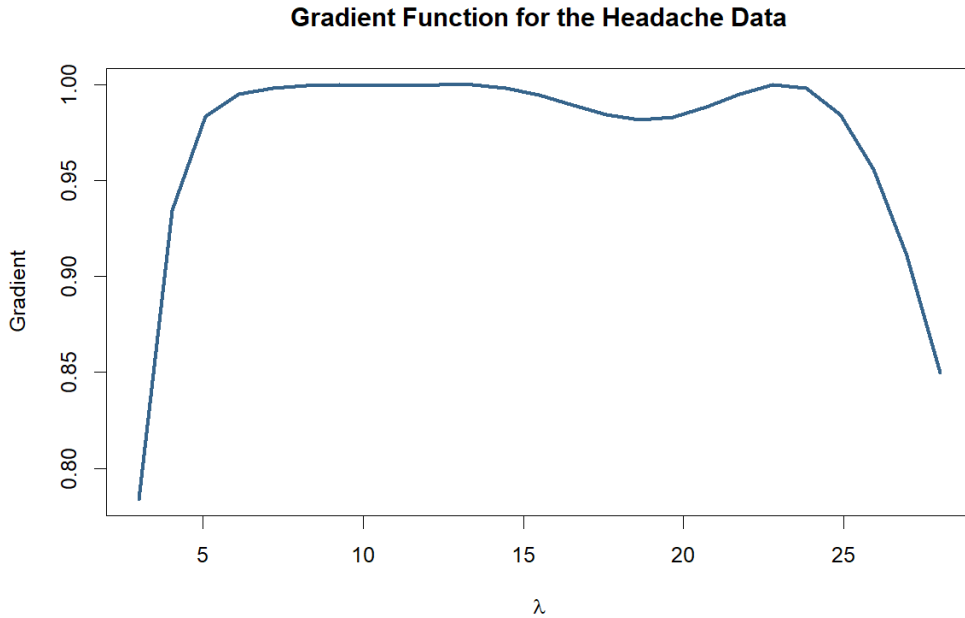


Figure 2: NPMLE gradient function for the baseline headache-frequency data.

The gradient function showed departures from a homogeneous one-component structure and suggested the presence of multiple regions in the underlying mixing distribution.

This provided exploratory support for fitting finite mixture models with more than one component. The number of components was subsequently assessed using fitted mixture models and information criteria.

5.5 Comparison of Poisson Finite Mixture Models

A single-component Poisson model was first fitted as a reference. Two-, three-, four-, and five-component Poisson finite mixture models were then fitted and compared using the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC).

Table 4: Comparison of Poisson and finite mixture models

Components	Log-Likelihood	AIC	BIC
1	-1476.032	2954.063	2958.057
2	-1309.058	2624.115	2636.097
3	-1306.585	2623.170	2643.140
4	-1306.585	2627.169	2655.127
5	-1306.548	2631.095	2667.041

The finite mixture models produced substantially lower AIC and BIC values than the single-component Poisson model. The three-component model had the lowest AIC (2623.170), whereas the two-component model had the lowest BIC (2636.097).

The four- and five-component models did not provide a sufficient improvement in model fit to compensate for their additional complexity. The three-component model was therefore retained for detailed interpretation because it provided the lowest AIC and yielded three distinct and interpretable headache-frequency groups.

The difference between the AIC and BIC selections is important. The results do not provide unanimous support for a single number of components. Instead, AIC favored three components, while the more strongly penalized BIC favored two components. The three-component solution was therefore examined in greater detail as the principal interpretable model.

5.6 Three-Component Poisson Mixture Model

The estimated parameters of the three-component Poisson mixture are presented in Table 5. The component labels reported by FlexMix are software-specific and are not ordered according to headache frequency.

Table 5: Estimated parameters of the three-component Poisson mixture

Frequency Group	λ	Proportion	Patients
Low-frequency	9.61	0.2512	102
Moderate-frequency	13.80	0.3729	148
High-frequency	22.95	0.3759	151

The estimated component means indicate substantial differences in baseline headache frequency. The low-frequency group had an estimated mean of 9.61 headache days and represented 25.12% of the sample. The moderate-frequency group had an estimated mean of 13.80 days and represented 37.29% of the sample. The high-frequency group had the highest estimated mean, 22.95 days, and represented 37.59% of the sample.

For interpretation, the three latent groups were therefore ordered from lowest to highest estimated headache frequency as low-frequency, moderate-frequency, and high-frequency groups.

The original FlexMix component labels are retained in the posterior classification plot because the plot represents the component assignments returned by the fitted model.

5.7 Comparison of CAMAN and FlexMix Estimates

The three-component solution was estimated independently using CAMAN and FlexMix. The resulting estimates are compared in Table 6. Because mixture-component labels are arbitrary and may differ between software implementations, the comparison is based on the ordering of the estimated component means rather than on the numerical component labels.

Table 6: Comparison of CAMAN and FlexMix estimates by headache-frequency group

Frequency Group	CAMAN λ	FlexMix λ	CAMAN Proportion	FlexMix Proportion
Low-frequency	9.140	9.608	0.191	0.251
Moderate-frequency	13.334	13.804	0.427	0.373
High-frequency	22.895	22.950	0.382	0.376

The component means estimated by the two approaches were similar. The absolute differences in the estimated means were 0.468, 0.470, and 0.055 headache days for the low-, moderate-, and high-frequency groups, respectively.

The estimated mixing proportions showed somewhat larger differences, particularly for the low- and moderate-frequency groups. Nevertheless, both approaches identified a similar three-level structure in the headache-frequency distribution.

The close agreement between the estimated component means provides support for the stability of the identified latent frequency groups across the two R implementations.

5.8 Posterior Classification of Patients

Posterior probabilities from the three-component FlexMix model were used to assign each patient to the component with the highest posterior probability. The resulting classification separated the patients into three groups corresponding to low-, moderate-, and high-frequency headache patterns.

The posterior classification is presented in Figure 3.

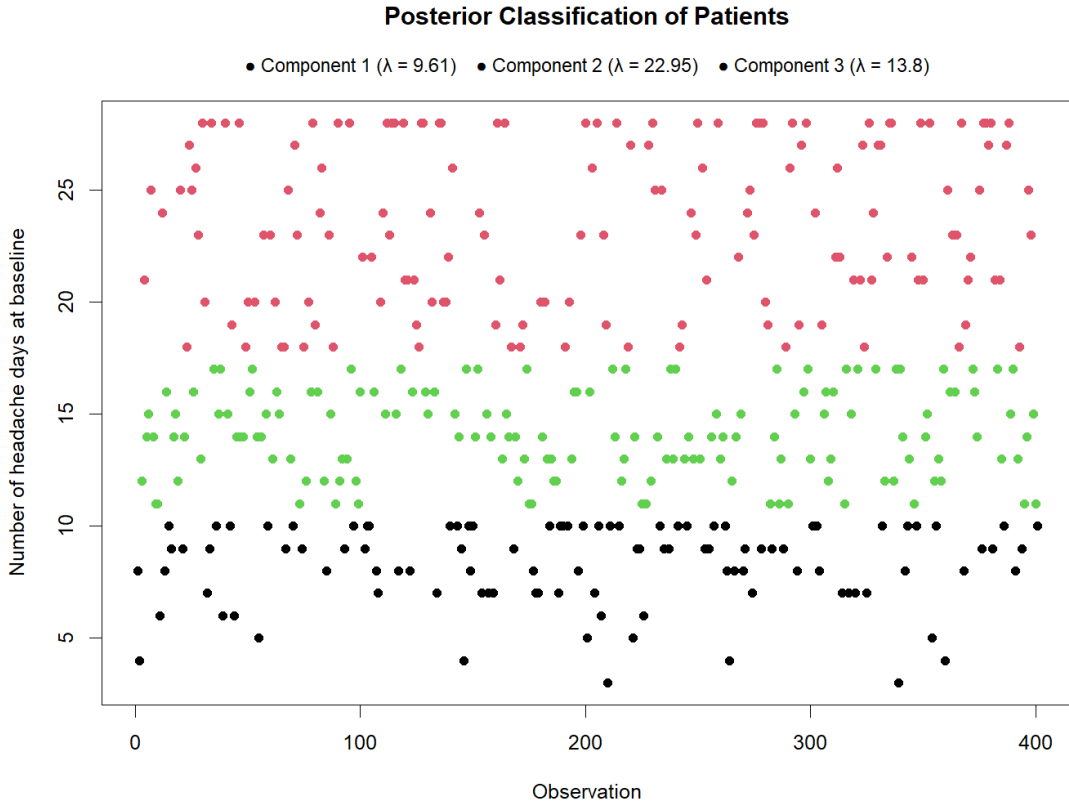


Figure 3: Posterior classification of patients under the three-component Poisson finite mixture model. Points are coloured according to the most likely latent component, with the estimated component means displayed below the title.

The classification plot shows a clear separation of the three estimated frequency groups. Patients assigned to the low-frequency group were concentrated at lower headache frequencies, whereas the high-frequency group contained patients with substantially more headache days. The moderate-frequency group occupied an intermediate range.

5.9 Assessment of the Final Three-Component Model

The fitted three-component model was further assessed by comparing observed and expected frequencies across the headache-frequency range.

Table 7: Selected observed and expected frequencies under the three-component model

Frequency	Observed	Expected	Pearson Residual
3	2	1.067	0.903
6	5	8.853	-1.295
10	33	23.159	2.045
15	21	20.615	0.085
20	13	15.030	-0.524
25	10	11.276	-0.380
27	10	8.311	0.586
28	38	6.785	11.984

For most of the displayed frequencies, the observed and expected counts were reasonably close. The main discrepancy occurred at the upper boundary of 28 headache days. Thirty-eight patients were observed at this value, compared with approximately 6.79 expected under the fitted mixture. The resulting Pearson residual was 11.98.

This indicates substantial lack of fit at the upper boundary. Thus, while the mixture model captured the main heterogeneous structure of the data, it did not fully reproduce the concentration of patients reporting the maximum observed number of headache days.

5.10 Three-Component Model Including Age

A three-component Poisson mixture model including age as a covariate was subsequently fitted using SAS. The model had an AIC of 2612.2 and a BIC of 2644.2.

Table 8: Age effects in the three-component Poisson mixture model

Component	Estimate	Standard Error	<i>p</i> -value
1	0.002208	0.001975	0.2637
2	-0.024140	0.005593	< 0.0001
3	0.003479	0.004569	0.4464

Age was statistically significant in SAS Component 2 ($p < 0.0001$). The estimated coefficient was negative ($\beta = -0.02414$). On the log scale, this corresponds to a multiplicative change of

$$\exp(-0.02414) \approx 0.976.$$

Thus, within this component, a one-year increase in age was associated with approximately a 2.4% decrease in the expected headache frequency.

The age effects for Components 1 and 3 were not statistically significant. Their estimated coefficients were small and did not provide sufficient evidence of an association between age and headache frequency within those components.

Because latent-component labels can change between independently fitted mixture models, SAS Component 2 is not assumed to correspond directly to Component 2 in the earlier FlexMix model.

5.11 Summary of Results

The analysis showed substantial heterogeneity in baseline headache frequency among the 401 patients. The variance-to-mean ratio of 2.78 indicated clear over-dispersion relative to the standard Poisson model.

The NPMLE gradient analysis supported a heterogeneous mixing distribution, and the finite mixture models provided substantially lower information criteria than the single-component Poisson reference model. AIC favored the three-component model, while BIC favored two components. The three-component solution was examined further because it provided the lowest AIC and produced three interpretable headache-frequency groups.

The estimated mean headache frequencies for the three components were 9.61, 13.80, and 22.95 days, with the groups representing approximately 25%, 37%, and 38% of the sample. Posterior classification showed clear separation of patients across these frequency levels.

The comparison between CAMAN and FlexMix produced similar component-specific mean frequencies, although the estimated mixing proportions differed between the two approaches. The final model diagnostics showed that the mixture captured the broad structure of the data but substantially underestimated the number of patients reporting 28 headache days.

The age-adjusted analysis showed that the association between age and headache frequency was not uniform across the latent population. Age was statistically significant in one SAS component, with a negative coefficient, whereas the effects in the other two components were not statistically significant.

6 Discussion

The aim of this study was to investigate heterogeneity in baseline headache frequency and to assess whether a Poisson finite mixture model could provide a more informative representation of the patient population than a single Poisson distribution. The analysis provided evidence of substantial variation in headache frequency and supported the presence of more than one underlying frequency level among the patients.

6.1 Heterogeneity in Headache Frequency

The initial descriptive analysis showed considerable variability in the number of headache days reported by patients. The variance of 44.98 was substantially larger than the mean of 16.18, giving a variance-to-mean ratio of 2.78. This is inconsistent with the equidispersion property of a standard Poisson distribution.

The over-dispersion observed in these data may reflect differences between patients in their underlying susceptibility to headaches. Rather than treating the additional variation only as a statistical problem, the finite mixture approach allowed the analysis to investigate whether the population could be represented by several latent groups with different headache-frequency rates.

This interpretation is consistent with the purpose of finite mixture models, which allow an observed population to be represented as a combination of unobserved subpopulations with different parameter values [5].

6.2 Evidence from the NPMLE Analysis

The NPMLE analysis provided an exploratory assessment of the underlying mixing distribution. The gradient function showed departures from a homogeneous one-component distribution and suggested that more than one support point may be present.

The NPMLE was particularly useful because it allowed the underlying mixing distribution to be explored before fixing the number of components in the finite mixture models. This is consistent with the role of NPMLE in mixture modelling, where the mixing distribution can be estimated without imposing a specific finite number of support points in advance [3, 4].

The gradient analysis should, however, be interpreted as exploratory evidence rather than as a definitive test for the exact number of components. The final choice of the finite mixture structure was therefore based on the fitted models, information criteria, and the interpretability of the estimated groups.

6.3 Choice of the Number of Components

The comparison of the candidate models demonstrated the importance of considering both model fit and model complexity. The single Poisson model performed substantially worse than the finite mixture models, providing strong evidence that a homogeneous Poisson model was inadequate for these data.

Among the mixture models, AIC reached its minimum for the three-component model, whereas BIC reached its minimum for the two-component model. This difference reflects the stronger penalty for model complexity imposed by BIC [1, 7].

The results therefore do not provide unanimous statistical evidence for exactly three components. Nevertheless, the three-component model was retained as the principal model for interpretation because it provided the lowest AIC and produced three clearly separated and interpretable headache-frequency levels.

This decision should be viewed as a modelling choice rather than as evidence that the population necessarily contains exactly three biologically distinct groups. The latent components are statistical representations of heterogeneity in the observed distribution.

6.4 Interpretation of the Three Latent Groups

The estimated component means from the three-component model were 9.61, 13.80, and 22.95 headache days. These estimates indicate substantial differences in the underlying headache-frequency rates.

The lowest-frequency component represented approximately one quarter of the sample, while the intermediate- and high-frequency components each represented approximately two-fifths of the population. The high-frequency group therefore represented a substantial proportion of the patients rather than a small group of extreme observations.

The posterior classification plot also showed a clear separation of patients across the three estimated frequency levels. This supports the interpretation that the mixture model is capturing meaningful heterogeneity in the observed distribution.

However, the latent groups should not be interpreted as fixed clinical categories without additional clinical information. They are statistical groups defined by differences in the estimated headache-frequency distributions.

6.5 Agreement Between CAMAN and FlexMix

The analysis used two different approaches to investigate the mixture structure. CAMAN was used for the NPMLE analysis, while FlexMix was used to estimate and classify patients under the three-component finite mixture model.

The estimated component means obtained from the two approaches were similar, particularly for the highest-frequency component. Although the estimated mixing pro-

portions differed to some extent, both methods identified a comparable set of low-, intermediate-, and high-frequency rates.

This agreement provides additional support for the existence of heterogeneity in the headache-frequency distribution. Differences between the estimated proportions are not unexpected because the methods use different estimation procedures and because mixture components are latent rather than directly observed.

It is also important to recognize that component labels are arbitrary in finite mixture models. Therefore, Component 1 in one software implementation does not necessarily correspond to Component 1 in another implementation. Comparisons should instead be based on the estimated component parameters and their ordering.

6.6 Model Adequacy and the Upper Boundary

Although the three-component model captured the major structure of the observed distribution, the model diagnostics revealed an important limitation. The number of patients reporting the maximum observed value of 28 headache days was substantially larger than expected under the fitted mixture.

Thirty-eight patients were observed at 28 headache days, compared with approximately 6.79 expected under the three-component model. The corresponding Pearson residual was 11.98.

This concentration at the upper boundary suggests that the observed distribution contains a feature that is not fully represented by the ordinary Poisson mixture. One possible explanation is that the four-week observation period imposes a natural upper limit of 28 headache days. Patients who experience headaches throughout the entire observation period are therefore concentrated at the boundary rather than continuing along an unrestricted count distribution.

This feature is important because the standard Poisson mixture treats the count distribution as though its support were not constrained by such a maximum. A model that explicitly accounts for a bounded or zero/upper-inflated outcome could potentially provide a better description of this aspect of the data.

The finding does not invalidate the mixture analysis. Instead, it shows that the finite mixture model captures the broad latent heterogeneity while leaving a specific boundary feature unexplained.

6.7 Effect of Age

The age-adjusted three-component model provided evidence that the relationship between age and headache frequency was not uniform across the latent population.

Age was statistically significant in one of the SAS mixture components, with an estimated coefficient of -0.02414 . On the original rate scale, this corresponds to a multi-

plicative factor of approximately 0.976 for a one-year increase in age. Thus, within that component, the expected headache frequency decreased by approximately 2.4% for each additional year of age.

The age effects in the other two components were not statistically significant. This suggests that the association between age and headache frequency may differ across latent groups rather than following a single population-wide effect.

The component-specific nature of this result is important. It should not be interpreted as evidence that increasing age generally reduces headache frequency among all patients. Rather, the result indicates that age was associated with the expected headache rate within one particular latent component.

6.8 Strengths of the Analysis

A major strength of the analysis was the use of several complementary methods rather than relying on a single fitted model. The descriptive analysis identified over-dispersion, the NPMLE provided an exploratory assessment of the mixing distribution, and finite mixture models were then compared using information criteria.

The use of both CAMAN and FlexMix also provided an opportunity to examine whether similar latent structures could be obtained using different estimation approaches. The broadly similar component means obtained from the two methods strengthen the interpretation of the identified heterogeneity.

The analysis was also implemented in both R and SAS. This provided an additional check on the model estimates and allowed the age-adjusted mixture model to be examined using SAS.

6.9 Limitations

Several limitations should be considered when interpreting the results.

First, the finite mixture components are latent statistical subpopulations and should not automatically be interpreted as distinct clinical disease categories. Additional clinical variables would be required to establish whether the identified groups correspond to meaningful clinical phenotypes.

Second, the AIC and BIC results did not select the same number of components. AIC favored three components, whereas BIC favored two. The three-component model was therefore selected as the principal model for interpretation, but the two-component solution remains a plausible alternative.

Third, the model showed substantial lack of fit at the upper boundary of 28 headache days. This indicates that the ordinary Poisson mixture does not fully account for the bounded nature of the outcome.

Finally, the analysis focused primarily on baseline headache frequency and age. Other patient characteristics may also contribute to the observed heterogeneity and could be considered in future models.

6.10 Implications for Further Analysis

The results suggest several directions for further statistical investigation. A model that explicitly accounts for the upper boundary of 28 headache days could be considered to address the concentration of observations at the maximum value.

Additional covariates could also be incorporated into the mixture model to determine whether demographic or clinical characteristics help explain membership in the latent frequency groups.

Future work could further compare alternative count distributions and mixture specifications using both statistical fit and substantive interpretability. Such analyses would help determine whether the three-component Poisson mixture provides the most useful description of the underlying patient heterogeneity or whether a more flexible model is warranted.

6.11 Overall Interpretation

Taken together, the findings indicate that baseline headache frequency was heterogeneous across the study population and that a single Poisson distribution was insufficient to describe the observed variation.

The finite mixture analysis provided evidence for several underlying headache-frequency levels, with the three-component model identifying low-, intermediate-, and high-frequency groups. Although the three-component model was not preferred by every information criterion, it provided the lowest AIC and produced an interpretable latent structure.

At the same time, the substantial lack of fit at 28 headache days shows that the mixture model does not explain every feature of the observed distribution. The findings therefore support the use of finite mixture modelling as a useful representation of heterogeneity in these data, while also indicating areas where a more specialized count model may be required.

7 Conclusion

This study investigated the distribution of baseline headache frequency using Poisson finite mixture models. The analysis began with a standard Poisson model and then considered finite mixtures to account for the substantial heterogeneity observed among patients.

The exploratory analysis showed clear over-dispersion, with a variance-to-mean ratio of 2.78. The NPMLE analysis provided further evidence that the observed population was unlikely to be adequately represented by a single homogeneous Poisson distribution.

Finite mixture models with two, three, four, and five components were compared. Although the Bayesian Information Criterion favored the two-component model, the Akaike Information Criterion favored the three-component model. The three-component solution was therefore selected for detailed interpretation because it provided the lowest AIC and produced a clear and interpretable separation of the population into low-, intermediate-, and high-frequency headache groups.

The estimated component means were approximately 9.61, 13.80, and 22.95 headache days, demonstrating substantial differences in underlying headache-frequency rates between the latent groups. Results obtained using CAMAN and FlexMix showed broadly similar component means, providing additional support for the identified heterogeneity.

The analysis also demonstrated an important limitation of the fitted Poisson mixture. The model substantially underestimated the number of patients reporting the maximum value of 28 headache days. This suggests that the bounded nature of the four-week observation period should be considered in future modelling.

Including age as a covariate showed that its association with headache frequency was not uniform across the latent groups. Age was statistically significant in one component, where a one-year increase in age was associated with an estimated 2.4% decrease in the expected headache frequency, while no significant age effects were identified in the other two components.

Overall, the findings demonstrate that Poisson finite mixture models provide a useful framework for investigating unobserved heterogeneity in headache-frequency data. The analysis also illustrates the importance of combining exploratory methods, model-selection criteria, graphical diagnostics, and substantive interpretation when choosing and evaluating a finite mixture model.

Future work could consider models that explicitly account for the upper boundary of the headache-frequency outcome and could incorporate additional clinical or demographic covariates to further explain the observed heterogeneity.

References

- [1] Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. doi: 10.1109/TAC.1974.1100705.
- [2] Alcinei Místico Azevedo, Ítallo Jesus Silva, Marcela Carlota Nery, Honovan Paz Rocha, and Rogério Alves Santana. Counting models for overdispersed data: A review with application to tuberculosis data. *Brazilian Journal of Biometrics*, 41(3): 274–286, 2023. doi: 10.28951/bjb.v41i3.635.
- [3] Bruce G. Lindsay. The geometry of mixture likelihoods: A general theory. *The Annals of Statistics*, 11(1):86–94, 1983. doi: 10.1214/aos/1176346059.
- [4] Bruce G. Lindsay. The geometry of mixture likelihoods, part ii: The exponential family. *The Annals of Statistics*, 11(3):783–792, 1983. doi: 10.1214/aos/1176346245.
- [5] Geoffrey J. McLachlan and David Peel. *Finite Mixture Models*. John Wiley & Sons, New York, 2000. doi: 10.1002/0471721182.
- [6] Geert Molenberghs and Geert Verbeke. Advanced modeling techniques: Course notes, 2017. Course notes.
- [7] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978. doi: 10.1214/aos/1176344136.
- [8] Kimberly F. Sellers, Sharad Borle, and Galit Shmueli. The com-poisson model for count data: A survey of methods and applications. *Applied Stochastic Models in Business and Industry*, 28(2):104–116, 2012. doi: 10.1002/asmb.918.
- [9] Andrew J. Vickers, Rebecca W. Rees, Catherine E. Zollman, Rob McCarney, Claire M. Smith, Nadia Ellis, Peter Fisher, and Robbert Van Haselen. Acupuncture for chronic headache in primary care: large, pragmatic, randomised trial. *BMJ*, 328(7442):744, 2004. doi: 10.1136/bmj.38029.421863.EB.